REBAL | Non-image experiments

From-scratch tabular and text representation learning
Experimental report - 12 October 2026

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Completed and independently checked: 15 condition-by-seed cells, 45 independently trained trunks, 135 retrained linear heads, and 405 named readout evaluations across Covertype and two DBpedia imbalance conditions. Every trunk completed 50 epochs; every retrained head received 2,000 updates. The 405 evaluations include 90 uniform-prior LA aliases that duplicate their base head predictions.

Measured primary effects

Condition	SMOTE minus ROS	CB+EQ minus CB
Covertype	+0.15 pp [+0.12, +0.17]	-0.04 pp [-0.20, +0.12]
DBpedia 10:1	+0.00 pp [-0.01, +0.02]	+0.00 pp [-0.01, +0.02]
DBpedia 100:1	+0.01 pp [-0.05, +0.06]	-0.02 pp [-0.04, +0.01]

Effects are paired differences in macro-F1, in percentage points, with unadjusted Student-t 95% intervals across five seeds. The first contrast holds the learned CB+EQ representation fixed and changes only the classifier sampler. The second compares equalization with its matched class-balanced control using natural classifier retraining.

Interpretation

The observed interpolation effect is Covertype: +0.15 points, DBpedia 10:1: +0.00 points, DBpedia 100:1: +0.01 points. The complete method tables and all six prespecified paired comparisons follow. Negative and small effects are retained.

Scope: these are from-scratch MLP and neural bag-of-words adaptations. They test latent classifier rebalancing and equalization under fixed data and compute protocols. They do not reproduce the image architecture, test a generator, establish transformer performance, or provide an optimized tabular state-of-the-art comparison.

Five seeds limit inference. The smallest attainable two-sided exact sign-flip p-value is 0.0625 before correction. Numerical consistency is independently verified; that verification does not imply that any method is superior.

1. Data, source scale, and separation

Condition	Source rows	Fitted	Calibration	Final test
Covertype	581,012	406,704	87,150	87,158
DBpedia 10:1	630,000	202,718	22,572	70,000
DBpedia 100:1	630,000	119,551	13,310	70,000

Covertype has 54 predictors and seven labels [1]. Its fixed stratified 70/15/15 allocation is a newly specified large-training protocol, not the historical benchmark partition. The ten numeric predictors use fit-only standardization; 44 binary indicators remain unchanged. The verified source has no exact predictor duplicates or conflicting-label groups.

DBpedia has 560,000 native training and 70,000 native test documents across 14 classes [2,3]. Native test remains intact with 5,000 documents per class. Connected duplicate components use normalized full title/content and nonempty content-only keys. Development exclusions remove 19 rows matching native test and 121 redundant rows, with no label conflicts, leaving 559,860 canonical development rows. Test labels do not determine exclusions.

A candidate holdout reserves 4,000 DBpedia documents per class. A common feasible maximum of 35,915 fitting documents receives rank-dependent long-tail thinning, with the same thinning fractions applied to calibration. This produces 3,591-35,915 fit examples/class at nominal ratio 10 and 359-35,915 at ratio 100. Calibration tails retain 400 and 40 examples/class, respectively. Exact duplicate checks do not establish paraphrase or entity independence.

Fixed split and rank identities

Split seed 1729 and DBpedia rank seed 2026 remain fixed across optimization seeds 0-4. Covertype ranks follow fitted class frequency. DBpedia conditions use a common semantic rank assignment and nested thinning. Head/middle/tail groups contain 2/2/3 Covertype classes and 4/5/5 DBpedia classes. These are rank groups, not universal few-shot definitions.

No scaler, vocabulary, neighbor graph or class prior is fitted on calibration/test rows. Primary runs omit ECOD. All original-row identities, actual class counts, source hashes, exclusions and license information are retained in prepared manifests. The source size is not the number of examples fitted per condition.

2. Architecture and matched controls

Component	Prespecified behavior
Tabular representation	MLP 54 -> 256 -> 256 -> 128 -> logits; ReLU and dropout 0.1.
Text representation	Random 128-D word embeddings; mean pooling -> 256 -> 128 -> logits. Fit-only 50,000-word vocabulary, 256-token cap, explicit PAD/UNK/SEP.
Three trunks	CE, deferred class-balanced CE, and deferred class-balanced CE plus equalization. Same initialization and natural batch order.
Trunk optimization	50 epochs; batch 1,024; LR 0.05; 5 warmup epochs then cosine decay; Nesterov SGD momentum 0.9; linear L2 0.0005; per-tensor norm clip 1.
Loss adaptations	CB/EQ activate after 80% of epochs. beta=1-1/n_max; fit-example expected weight=1. EQ coefficient 0.05.
Classifier retraining	Natural, class-uniform ROS, streaming latent SMOTE; same initialization, 2,000 updates, batch 2,048; SGD LR 0.01, momentum 0.9, L2 0.0005.
Readouts	3 native, 3 NCM, 9 retrained heads, 12 LA variants = 27 readouts per cell. LA/NCM are not additional encoders.

The count-aware CB beta and fit-example weight normalization are explicit large-data adaptations of effective-number weighting [5]. They prevent large class counts from making the source beta=0.999 nearly uniform. The isolated EQ contrast is CB+EQ versus CB; this three-trunk design is not a complete two-factor loss ablation.

The repository's geometric EQ normalizes feature vectors, minimizes the mean within-class squared radius, and subtracts the mean pairwise squared distance of eligible class centers. A class needs at least two batch examples; between-class separation requires at least two eligible classes. This specific regularizer is an implementation under test, not a claim established by the cited loss papers.

Following representation/classifier separation [6], streaming SMOTE adapts within-class interpolation [4]: choose a class uniformly, then a real anchor and an exact within-class neighbor. A fresh interpolation is drawn with probability 1-n_class/n_max. Maximal classes use real samples only. This differs from a finite static synthetic pool. It interpolates learned vectors, never token IDs, one-hot categories or physical measurements.

Post-hoc LA [7] uses each head's effective training prior. Balanced ROS/SMOTE priors are uniform, so tau is zero and predictions do not change. The native CB/REBAL prior approximates the final weighted objective after deferred training. Other tau values use calibration macro-F1 on the fixed grid. The final epoch is evaluated; test scores select no checkpoint. The embedding-average text model is related to [8], but adds nonlinear layers and does not reproduce fastText.

3. Complete results: Covertype | natural imbalance

Mean +/- sample standard deviation across five optimization seeds. Values are percentages. Macro recall is balanced accuracy; it equals accuracy on the balanced DBpedia test. Zero-denominator class metrics use zero.

Method / readout	Accuracy	Macro F1	Macro recall	Tail F1
CE	89.33 +/- 0.10	76.68 +/- 0.55	72.44 +/- 0.58	60.52 +/- 1.11
CE + LA	89.17 +/- 0.14	78.57 +/- 0.64	78.72 +/- 0.74	65.43 +/- 1.22
CE / NCM	62.58 +/- 0.73	52.53 +/- 0.71	75.99 +/- 0.63	36.39 +/- 0.91
CE / natural	89.23 +/- 0.12	75.58 +/- 1.14	71.00 +/- 1.21	58.07 +/- 2.48
CE / natural + LA	89.16 +/- 0.13	78.30 +/- 0.78	77.96 +/- 0.95	64.81 +/- 1.54
CE / ROS	83.41 +/- 0.24	68.85 +/- 0.61	84.81 +/- 0.54	50.42 +/- 1.05
CE / ROS + LA	83.41 +/- 0.24	68.85 +/- 0.61	84.81 +/- 0.54	50.42 +/- 1.05
CE / SMOTE	83.50 +/- 0.23	68.95 +/- 0.59	84.78 +/- 0.53	50.55 +/- 1.02
CE / SMOTE + LA	83.50 +/- 0.23	68.95 +/- 0.59	84.78 +/- 0.53	50.55 +/- 1.02
CB	85.31 +/- 0.12	72.70 +/- 0.39	87.37 +/- 0.33	56.95 +/- 0.71
CB + LA	85.31 +/- 0.12	72.70 +/- 0.39	87.37 +/- 0.33	56.95 +/- 0.71
CB / NCM	61.01 +/- 0.62	51.86 +/- 0.67	76.74 +/- 0.63	36.97 +/- 0.88
CB / natural	88.85 +/- 0.16	77.19 +/- 1.23	73.75 +/- 1.42	61.93 +/- 2.59
CB / natural + LA	88.72 +/- 0.10	79.46 +/- 0.64	81.26 +/- 0.86	67.75 +/- 1.30
CB / ROS	83.75 +/- 0.18	71.09 +/- 0.53	87.08 +/- 0.26	55.04 +/- 0.99
CB / ROS + LA	83.75 +/- 0.18	71.09 +/- 0.53	87.08 +/- 0.26	55.04 +/- 0.99
CB / SMOTE	83.84 +/- 0.19	71.23 +/- 0.54	87.08 +/- 0.27	55.27 +/- 1.01
CB / SMOTE + LA	83.84 +/- 0.19	71.23 +/- 0.54	87.08 +/- 0.27	55.27 +/- 1.01
CB+EQ	85.23 +/- 0.12	72.58 +/- 0.39	87.35 +/- 0.34	56.78 +/- 0.70
CB+EQ + LA	85.23 +/- 0.12	72.58 +/- 0.39	87.35 +/- 0.34	56.78 +/- 0.70
CB+EQ / NCM	60.34 +/- 0.62	51.31 +/- 0.71	76.51 +/- 0.63	36.60 +/- 1.01
CB+EQ / natural	88.84 +/- 0.15	77.15 +/- 1.27	73.78 +/- 1.47	61.90 +/- 2.70
CB+EQ / natural + LA	88.66 +/- 0.09	79.38 +/- 0.63	81.35 +/- 0.87	67.68 +/- 1.29
CB+EQ / ROS	83.62 +/- 0.18	70.93 +/- 0.54	87.02 +/- 0.27	54.85 +/- 1.00
CB+EQ / ROS + LA	83.62 +/- 0.18	70.93 +/- 0.54	87.02 +/- 0.27	54.85 +/- 1.00
CB+EQ / SMOTE	83.72 +/- 0.18	71.08 +/- 0.55	87.01 +/- 0.29	55.09 +/- 1.01
CB+EQ / SMOTE + LA	83.72 +/- 0.18	71.08 +/- 0.55	87.01 +/- 0.29	55.09 +/- 1.01

CB+EQ denotes the REBAL trunk. '/ natural', '/ ROS' and '/ SMOTE' denote retrained linear heads on that trunk's frozen representation. NCM uses normalized fit centroids. LA variants reuse their corresponding head; duplicate rows for balanced priors are expected. Head/middle region metrics and all per-class scores are retained in the CSV evidence.

4. Complete results: DBpedia | nominal 10:1

Mean +/- sample standard deviation across five optimization seeds. Values are percentages. Macro recall is balanced accuracy; it equals accuracy on the balanced DBpedia test. Zero-denominator class metrics use zero.

Method / readout	Accuracy	Macro F1	Macro recall	Tail F1
CE	94.67 +/- 0.70	94.67 +/- 0.71	94.67 +/- 0.70	93.89 +/- 1.26
CE + LA	95.00 +/- 0.61	95.00 +/- 0.62	95.00 +/- 0.61	94.33 +/- 1.11
CE / NCM	95.37 +/- 0.70	95.36 +/- 0.70	95.37 +/- 0.70	94.92 +/- 1.21
CE / natural	94.55 +/- 0.74	94.54 +/- 0.75	94.55 +/- 0.74	93.69 +/- 1.31
CE / natural + LA	94.92 +/- 0.70	94.91 +/- 0.70	94.92 +/- 0.70	94.18 +/- 1.16
CE / ROS	95.64 +/- 0.66	95.65 +/- 0.65	95.64 +/- 0.66	95.13 +/- 1.14
CE / ROS + LA	95.64 +/- 0.66	95.65 +/- 0.65	95.64 +/- 0.66	95.13 +/- 1.14
CE / SMOTE	95.65 +/- 0.65	95.66 +/- 0.65	95.65 +/- 0.65	95.13 +/- 1.14
CE / SMOTE + LA	95.65 +/- 0.65	95.66 +/- 0.65	95.65 +/- 0.65	95.13 +/- 1.14
CB	95.71 +/- 0.62	95.72 +/- 0.61	95.71 +/- 0.62	95.30 +/- 1.06
CB + LA	95.71 +/- 0.62	95.72 +/- 0.61	95.71 +/- 0.62	95.30 +/- 1.06
CB / NCM	95.46 +/- 0.67	95.45 +/- 0.67	95.46 +/- 0.67	95.10 +/- 1.12
CB / natural	94.68 +/- 0.67	94.67 +/- 0.68	94.68 +/- 0.67	93.94 +/- 1.15
CB / natural + LA	94.91 +/- 0.49	94.91 +/- 0.49	94.91 +/- 0.49	94.25 +/- 0.92
CB / ROS	95.71 +/- 0.62	95.72 +/- 0.62	95.71 +/- 0.62	95.29 +/- 1.06
CB / ROS + LA	95.71 +/- 0.62	95.72 +/- 0.62	95.71 +/- 0.62	95.29 +/- 1.06
CB / SMOTE	95.72 +/- 0.62	95.72 +/- 0.61	95.72 +/- 0.62	95.30 +/- 1.04
CB / SMOTE + LA	95.72 +/- 0.62	95.72 +/- 0.61	95.72 +/- 0.62	95.30 +/- 1.04
CB+EQ	95.71 +/- 0.62	95.72 +/- 0.61	95.71 +/- 0.62	95.29 +/- 1.05
CB+EQ + LA	95.72 +/- 0.62	95.72 +/- 0.61	95.72 +/- 0.62	95.30 +/- 1.06
CB+EQ / NCM	95.44 +/- 0.66	95.42 +/- 0.67	95.44 +/- 0.66	95.11 +/- 1.10
CB+EQ / natural	94.68 +/- 0.68	94.67 +/- 0.69	94.68 +/- 0.68	93.95 +/- 1.16
CB+EQ / natural + LA	94.92 +/- 0.56	94.92 +/- 0.56	94.92 +/- 0.56	94.26 +/- 0.95
CB+EQ / ROS	95.72 +/- 0.61	95.73 +/- 0.61	95.72 +/- 0.61	95.30 +/- 1.04
CB+EQ / ROS + LA	95.72 +/- 0.61	95.73 +/- 0.61	95.72 +/- 0.61	95.30 +/- 1.04
CB+EQ / SMOTE	95.72 +/- 0.61	95.73 +/- 0.60	95.72 +/- 0.61	95.31 +/- 1.03
CB+EQ / SMOTE + LA	95.72 +/- 0.61	95.73 +/- 0.60	95.72 +/- 0.61	95.31 +/- 1.03

CB+EQ denotes the REBAL trunk. '/ natural', '/ ROS' and '/ SMOTE' denote retrained linear heads on that trunk's frozen representation. NCM uses normalized fit centroids. LA variants reuse their corresponding head; duplicate rows for balanced priors are expected. Head/middle region metrics and all per-class scores are retained in the CSV evidence.

5. Complete results: DBpedia | nominal 100:1

Mean +/- sample standard deviation across five optimization seeds. Values are percentages. Macro recall is balanced accuracy; it equals accuracy on the balanced DBpedia test. Zero-denominator class metrics use zero.

Method / readout	Accuracy	Macro F1	Macro recall	Tail F1
CE	73.57 +/- 0.77	68.31 +/- 0.76	73.57 +/- 0.77	40.46 +/- 2.28
CE + LA	79.59 +/- 0.81	76.39 +/- 1.03	79.59 +/- 0.81	55.24 +/- 2.49
CE / NCM	82.48 +/- 0.98	82.09 +/- 1.02	82.48 +/- 0.98	69.13 +/- 2.18
CE / natural	71.33 +/- 0.97	65.23 +/- 1.15	71.33 +/- 0.97	34.11 +/- 3.16
CE / natural + LA	79.21 +/- 1.47	76.05 +/- 2.17	79.21 +/- 1.47	54.51 +/- 5.60
CE / ROS	84.26 +/- 1.22	84.25 +/- 1.23	84.26 +/- 1.22	73.63 +/- 2.63
CE / ROS + LA	84.26 +/- 1.22	84.25 +/- 1.23	84.26 +/- 1.22	73.63 +/- 2.63
CE / SMOTE	84.23 +/- 1.21	84.21 +/- 1.22	84.23 +/- 1.21	73.61 +/- 2.59
CE / SMOTE + LA	84.23 +/- 1.21	84.21 +/- 1.22	84.23 +/- 1.21	73.61 +/- 2.59
CB	85.17 +/- 1.25	84.91 +/- 1.39	85.17 +/- 1.25	75.04 +/- 2.60
CB + LA	85.17 +/- 1.25	84.91 +/- 1.39	85.17 +/- 1.25	75.04 +/- 2.60
CB / NCM	84.68 +/- 1.11	84.51 +/- 1.18	84.68 +/- 1.11	74.80 +/- 2.14
CB / natural	72.61 +/- 0.90	67.08 +/- 0.88	72.61 +/- 0.90	38.98 +/- 2.45
CB / natural + LA	80.81 +/- 1.34	78.25 +/- 1.90	80.81 +/- 1.34	60.18 +/- 4.62
CB / ROS	85.84 +/- 1.23	85.88 +/- 1.24	85.84 +/- 1.23	77.62 +/- 2.20
CB / ROS + LA	85.84 +/- 1.23	85.88 +/- 1.24	85.84 +/- 1.23	77.62 +/- 2.20
CB / SMOTE	85.87 +/- 1.22	85.90 +/- 1.23	85.87 +/- 1.22	77.66 +/- 2.17
CB / SMOTE + LA	85.87 +/- 1.22	85.90 +/- 1.23	85.87 +/- 1.22	77.66 +/- 2.17
CB+EQ	85.16 +/- 1.27	84.88 +/- 1.43	85.16 +/- 1.27	74.98 +/- 2.66
CB+EQ + LA	85.16 +/- 1.27	84.88 +/- 1.43	85.16 +/- 1.27	74.98 +/- 2.66
CB+EQ / NCM	84.74 +/- 1.10	84.56 +/- 1.17	84.74 +/- 1.10	74.93 +/- 2.11
CB+EQ / natural	72.60 +/- 0.89	67.06 +/- 0.88	72.60 +/- 0.89	38.96 +/- 2.44
CB+EQ / natural + LA	80.80 +/- 1.33	78.24 +/- 1.88	80.80 +/- 1.33	60.18 +/- 4.58
CB+EQ / ROS	85.86 +/- 1.23	85.89 +/- 1.24	85.86 +/- 1.23	77.65 +/- 2.18
CB+EQ / ROS + LA	85.86 +/- 1.23	85.89 +/- 1.24	85.86 +/- 1.23	77.65 +/- 2.18
CB+EQ / SMOTE	85.87 +/- 1.22	85.90 +/- 1.24	85.87 +/- 1.22	77.68 +/- 2.18
CB+EQ / SMOTE + LA	85.87 +/- 1.22	85.90 +/- 1.24	85.87 +/- 1.22	77.68 +/- 2.18

CB+EQ denotes the REBAL trunk. '/ natural', '/ ROS' and '/ SMOTE' denote retrained linear heads on that trunk's frozen representation. NCM uses normalized fit centroids. LA variants reuse their corresponding head; duplicate rows for balanced priors are expected. Head/middle region metrics and all per-class scores are retained in the CSV evidence.

6. Paired effects and uncertainty

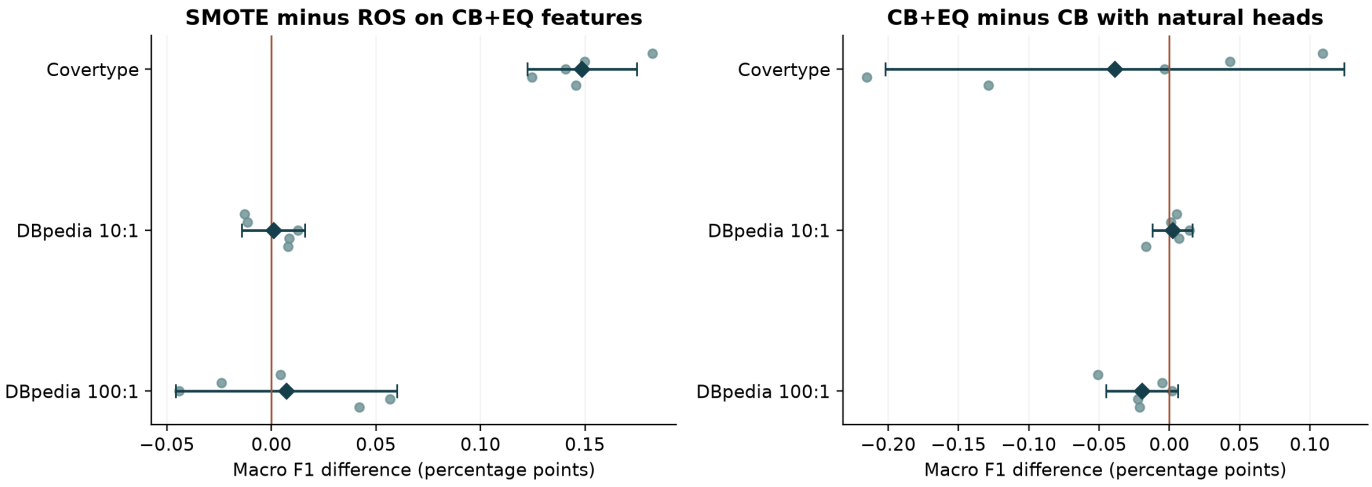


Figure 1. Points are individual paired seed differences. Diamonds and bars show the mean and unadjusted Student-t 95% interval. Positive values favor the first named method. The three conditions share two prespecified contrasts.

Condition / contrast	Delta (pp)	95% t interval	t p, Holm	Exact p, Holm
Covertype SMOTE - ROS	+0.15	[+0.12, +0.17]	5.75e-04	0.375
Covertype CB+EQ - CB	-0.04	[-0.20, +0.12]	1.000	1.000
DBpedia 10:1 SMOTE - ROS	+0.00	[-0.01, +0.02]	1.000	1.000
DBpedia 10:1 CB+EQ - CB	+0.00	[-0.01, +0.02]	1.000	1.000
DBpedia 100:1 SMOTE - ROS	+0.01	[-0.05, +0.06]	1.000	1.000
DBpedia 100:1 CB+EQ - CB	-0.02	[-0.04, +0.01]	0.498	0.625

Holm adjustment covers the complete six-comparison family, separately for the t and exact sign-flip tests. The displayed intervals are not multiplicity-adjusted. A five-seed t-test relies on distributional assumptions that are difficult to assess; exact sign-flip p-values cannot be below 0.0625. Nonsignificance is not equivalence. Shared evaluation rows do not create additional independent training replications.

7. Performance patterns and mechanism checks

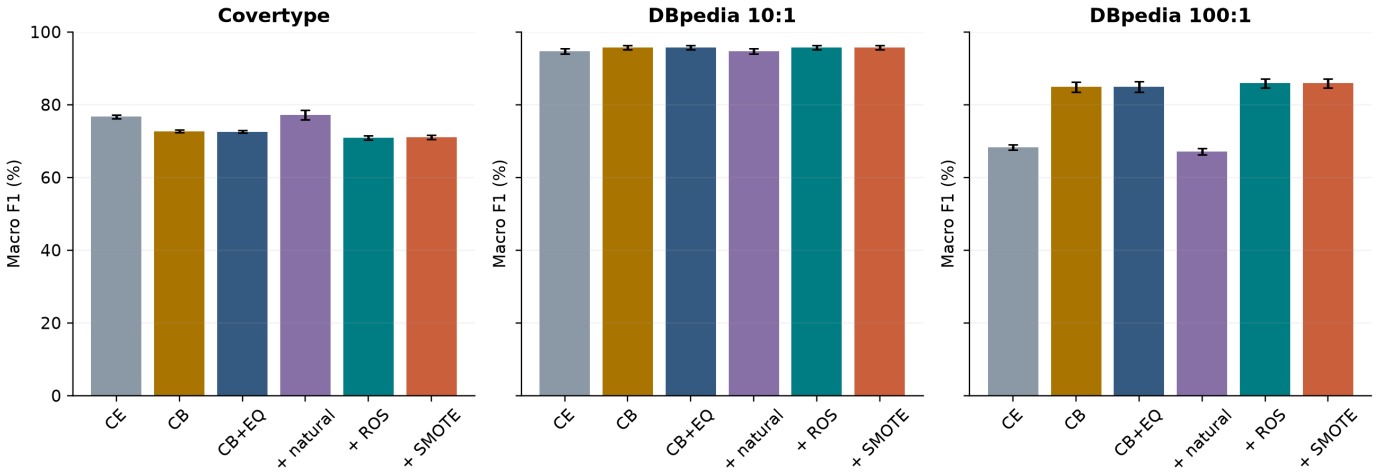


Figure 2. Macro-F1 for the three native trunks and the natural/ROS/SMOTE heads on CB+EQ representations. Bars show means; error bars show sample SD across five seeds. '+ natural', '+ ROS' and '+ SMOTE' share the CB+EQ representation. Every other method appears in the complete tables.

Equalization's measured exposure

Condition	Mean active-epoch eligibility	Smallest class-level fraction
Covertype	99.3%	94.9%
DBpedia 10:1	100.0%	100.0%
DBpedia 100:1	98.1%	80.4%

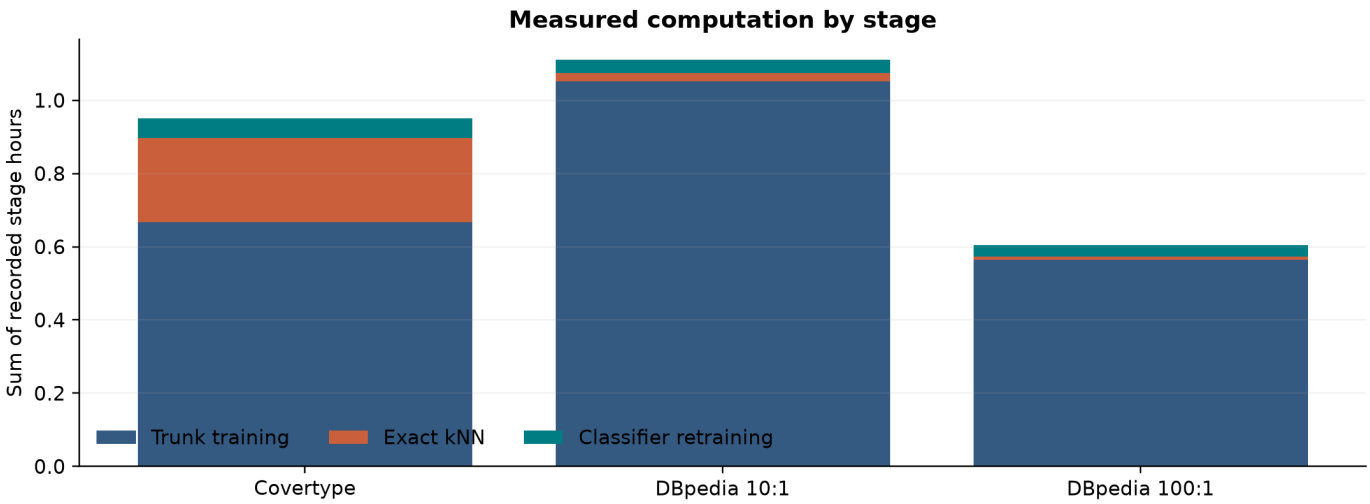
Eligibility is the fraction of active-EQ training batches containing at least two examples of a class, averaged as indicated. Low eligibility can limit the regularizer's influence on rare classes. This diagnostic is distinct from proving that equalization improved generalization.

Exploratory centroid readouts

CB+EQ-minus-CB NCM macro-F1 mean differences are Covertype: -0.55 points; DBpedia 10:1: -0.03 points; DBpedia 100:1: +0.05 points. NCM uses normalized fitted centroids and offers exploratory evidence about representation geometry. These comparisons are outside the six primary contrasts and support no additional confirmatory significance claim.

The report preserves negative outcomes and all controls. Effects are conditional on these learned representations, the chosen schedules and calibration populations. The highest isolated row is descriptive and does not replace the prespecified mechanism comparisons.

8. Actual execution and runtime



Condition	Trunk hours	kNN hours	Head hours
Covertype	0.666	0.230	0.053
DBpedia 10:1	1.051	0.023	0.036
DBpedia 100:1	0.565	0.009	0.031

Figure 3 and the table sum measured stage durations across all seeds and trunks. These sums are not elapsed wall time when workers overlap. They omit separate input preparation and feature extraction; lost work before an interrupted checkpoint may also be absent. Exact kNN remains quadratic within oversampled classes even though memory is blocked.

The observed span from study-plan creation to the last completed cell is 0.93 hours. It includes preparation, scheduling and possible idle time and should not be treated as isolated optimizer throughput.

python: 3.12.14; OS: Darwin; OS_release: 25.6.0; architecture: arm64; device: cpu; deterministic_algorithms: True; determinism_warn_only: True

Recorded software: numpy 2.5.4, scikit-learn 1.9.1, scipy 1.18.1, torch 2.14.1.

CPU and MPS train-only pilots were operational checks, not accuracy evaluations. Device choice preceded full testing. Seed control and deterministic-algorithm warnings do not guarantee bitwise equality across device families; EmbeddingBag reduction and floating-point neighbor ties are documented limits.

9. Evidence checks and reproducibility

An independent verifier recomputed classification metrics from saved logits and labels, checked every calibration curve, compared aggregate summaries and all six paired contrasts, and verified source, split, preprocessing, checkpoint and neighbor-graph artifact hashes. It did not import the runner's metric functions or load pickle checkpoints.

Evidence	Verified scope
Condition/seed cells	15 of 15; no missing cells
Readouts and calibration	405 readouts; 180 per-cell calibration curves
Aggregation	81 five-seed method summaries; complete six-contrast family
Training identity	Fixed source, protocol, data manifests, 50-epoch trunks and 2,000-update heads
Independent limits	Stored logits are verified, but model inference is not rerun. Graph/checkpoint bytes are hashed; nearest-distance optimality is not reconstructed.

The builder refused pilot, partial, different-budget or stale-verification evidence. It then checked that current aggregate, cell, prediction and artifact hashes still match the independently verified snapshot. Successful verification establishes consistency, not cryptographic authentication of an untrusted experiment or evidence of method superiority.

The independent verifier passed 11 portable synthetic and tamper tests. Reproduce those checks from the repository with `python -m unittest discover -s tests -p 'test_verify_nonimage.py' -v`. These tests exercise the checking software; the 15 completed study cells are separately verified.

Published code commit: `295d43f6d67746879cd0ef8bfa72f88b57821174`.

Training protocol fingerprint

`03e25e8a0dd6f9f1f7cd9db1daaba2850e04e2857d55cb6d0be5e41b241de0a5`

Runner source SHA256

`be58b2706400b85340736c0ca0e3f07cbcb23895c4d07e92a6789de86d072b21`

Independent verifier SHA256

`cdc93b840851a0e7f71d4c6075750c730d023a10de24d60431e1fcc900fbd4a8`

Verified aggregate SHA256

`aa7380f83410c4eb87f070e8d700f5695928c4f8fd1d0766da507bebbc8621bd`

Portable reproduction commands and full source identities are documented in docs/nonimage_experiment.md. The CSVs contain all seed-level scalar results, all per-class scores, summaries, six contrasts and recorded stage runtimes. CSV performance values are fractions; tables and plots use percentages.

10. Limits and references

Two data domains and fixed splits

Covertype can retain geographic dependence; DBpedia can retain paraphrases or related entities despite exact deduplication. Optimization seeds do not sample new datasets or train/test partitions. The two DBpedia ratios share source material and are correlated conditions.

A deliberately adapted model family

The tabular MLP and neural bag-of-words encoder are matched controls, not tuned state-of-the-art models. The study does not establish superiority over boosted trees or transformers. A fixed 50-epoch budget does not prove optimal convergence for every objective.

Resampling and calibration definitions

Streaming latent SMOTE redraws interpolations instead of training on one finite synthetic pool. ROS changes the sampling prior but does not create new information. Calibration is thinned by the same rank fractions for DBpedia; all adjustment claims remain conditional on those labeled calibration sets.

Loss and readout interpretation

Count-aware beta and expected-example weight normalization change the original large-class behavior. The native deferred CB/REBAL prior is an approximation. Balanced-head LA rows repeat their base predictions by design. The CB+EQ-versus-CB contrast isolates EQ only within this declared objective and schedule.

Uncertainty and negative findings

Five paired seeds limit exact-test resolution and make parametric assumptions hard to assess. Confidence intervals are conditional on these data, architectures and budgets. Small, inconsistent or negative differences must remain in the conclusions; no absent or failed run is imputed.

Primary references

1. Blackard (1998), UCI Covertype, DOI 10.24432/C50K5N
2. Zhang, Zhao and LeCun (2015), Character-level CNNs for Text Classification
3. DBpedia-14 maintained dataset card and source attribution
4. Chawla et al. (2002), SMOTE
5. Cui et al. (2019), Class-Balanced Loss
6. Kang et al. (2020), Decoupling Representation and Classifier
7. Menon et al. (2021), Long-tail Learning via Logit Adjustment
8. Joulin et al. (2017), Bag of Tricks for Efficient Text Classification (preprint 2016)

Covertype attribution and DBpedia CC BY-SA/GFDL terms accompany the prepared source manifests. Raw source records are not embedded in this report.